Supplementary Analysis 1: ELMo and BERT (large)

In the primary analyses reported above, we used Cosine Distance—a continuous measure of the distance between two contexts of use—derived from BERT-base (Devlin et al, 2018). BERT-base has 12 layers, 12 attention heads, and 110M parameters; we calculated Cosine Distance using embeddings from the final layer.

However, there are a number of other neural language models, which bear some important similarities and differences to BERT. One question that arises is to what extent the results reported above depend on which neural language model was used to calculate Cosine Distance. There are two important sub-components to this question. First, does a measure of Cosine Distance from other models also predict Response Time? Second, and more importantly, does using Cosine Distance from a more powerful model affect the predictive power of *other* covariates, such as Sense Boundary? The latter question is especially important, as it could have theoretical ramifications: if Sense Boundary no longer adds explanatory power beyond Cosine Distance, it would suggest that the Pure Exemplar Theory is correct.

Methods

To test this question, we considered two alternative language models: ELMo and BERT-large. ELMo (Peters et al, 2018) is a bidirectional LSTM, which has three layers, and also produces contextualized embeddings; it typically under-performs BERT in Word Sense Disambiguation tasks (Wiedemann et al, 2019). BERT-large, as the name implies, is simply a bigger version of BERT—it has 24 layers (compared to 12), 16 attention heads (compared to 12), and 336M parameters (compared to 110M).

Using both models, we extracted contextualized embeddings for the target word from each sentence pair, then calculated the Cosine Distance between these embeddings. We then conducted several analyses, designed to answer distinct questions. Each analysis used pooled data from Experiments 1-2.

Analysis 1: Assessing the predictive power of each model.

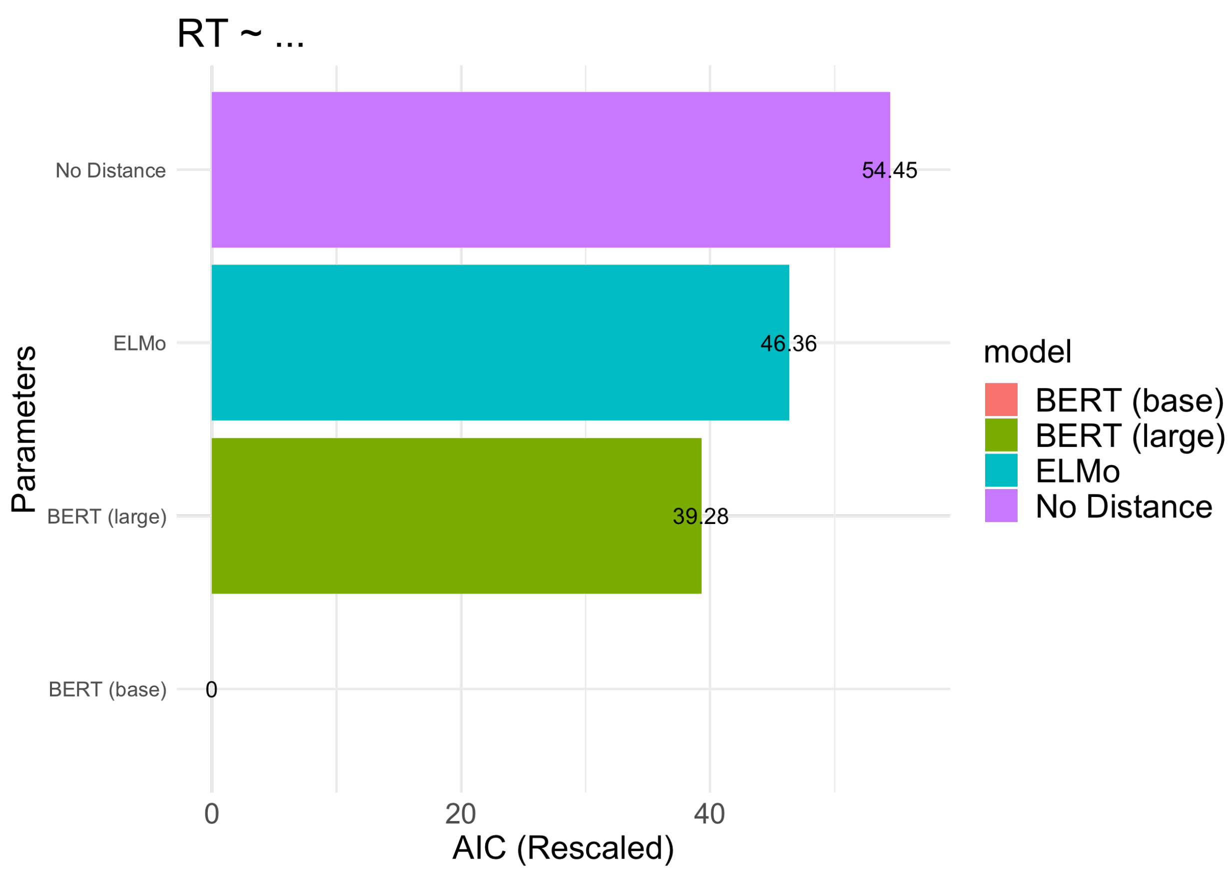
First, we asked: 1) whether each measure of Cosine Distance was predictive of human behavior (RT or Correct Response); and 2) which measure was most predictive. We built a series of mixed effects models, with either Log RT or Correct Response as a dependent variable, fixed effects of Cosine Distance (from either ELMo, BERT-base, or BERT-large), an interaction between Sense Boundary and Ambiguity Type (and main effects of each), by-subject random slopes for the effects of Sense Boundary and Ambiguity Type, and random intercepts for subjects, items, and experiment. We also constructed a reduced model omitting only the effect of Cosine Distance. Then, we compared each of the full models to this reduced model, and asked whether the full model explained significantly more variance.

For Log RT, we found that Cosine Distance measures from each of the three NLMs improved model fit: BERT-base [$X^2(1) = 56.45, p < .001$], BERT-large [$X^2(1) = 17.17, p < .001$], and ELMo [$X^2(1) = 10.09, p = .001$]. In other words, each NLM was able to explain significant variance in RT.

For Correct Response, only Cosine Distance from BERT-large improved model fit: [$X^2(1) = 11.71, p = .001$]. Recall that in the primary analyses, Cosine Distance from BERT-base did not explain significant variance in Correct Response; thus, this latter analysis suggests that the increase in model size (i.e., for BERT-large) does result in improved predictive power—at least when it comes to Correct Response.

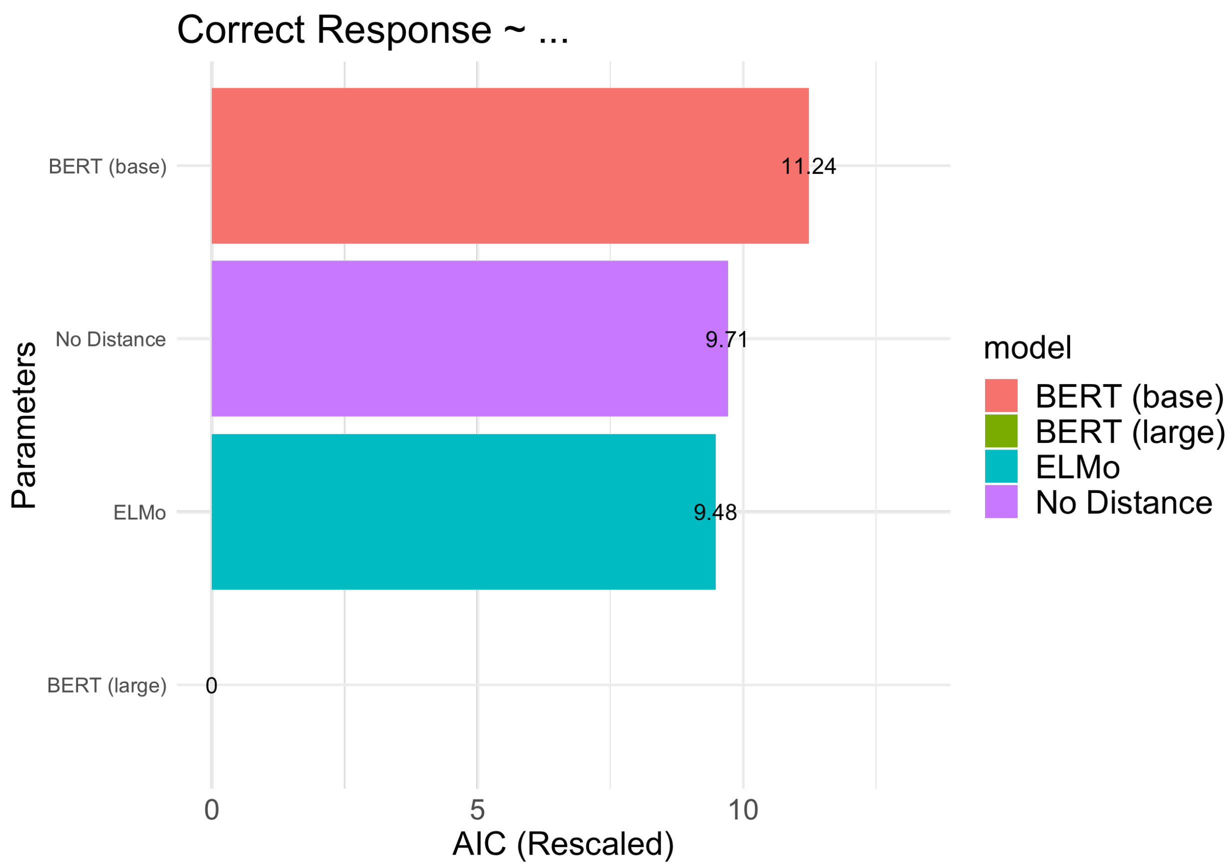
We also compared the AIC of each model, considering first the four models predicting RT, and then the four models predicting Correct Response. Within the set of models built for each dependent variable, AIC values were rescaled to the smallest AIC.

As depicted below, BERT-base exhibited the best fit (i.e., lowest AIC) when predicting RT. There was a considerable difference in predictive power between BERT-base and the second-most predictive model, BERT-large. This is somewhat surprising: BERT-large has many more parameters and twice as many layers, but is less predictive of response time for correct trials. On the other hand, it is consistent with other work (Kuribayashi et al, 2021) suggesting that improvements along some dimensions of model performance (e.g., lower perplexity) do not always entail better prediction of human behavior.



Supplementary Figure 1: Rescaled AIC for the four statistical models predicting RT across Experiments 1-2, using either Cosine Distance from one of three NLMs, or omitting Cosine Distance altogether. BERT-base exhibited the best fit (i.e., the lowest AIC).

On the other hand, *only* BERT-large was significantly predictive of Correct Response. Consistent with this result, BERT-large exhibited the lowest AIC when it came to predicting Correct Response, and BERT-base actually exhibited the *worst* fit—even lower than a model omitting Cosine Distance altogether.



Supplementary Figure 2: Rescaled AIC for the four statistical models predicting Correct Response across Experiments 1-2, using either Cosine Distance from one of three NLMs, or omitting Cosine Distance altogether. BERT-large exhibited the best fit (i.e., the lowest AIC).

Analysis 2: Identifying whether Sense Boundary remains predictive.

In the second analysis, we asked whether the main effect of Sense Boundary would still be observed if Cosine Distance measures from the other NLMs were included in the model. This represents a particularly stringent test of Sense Boundary—i.e., whether it explains independent variance from all three NLMs (ELMo, BERT-base, and BERT-large) combined. Obtaining a significant effect would act as another disconfirmation of Pure Exemplar Theory.

For both dependent variables (Correct Response and Log RT), we constructed a full model with three measures of Cosine Distance (from ELMo, BERT-base, and BERT-large), a fixed effect of Sense Boundary, by-subject random slopes for the effect of Sense Boundary, and random intercepts for subjects, experiment, and item.

We found that the addition of Sense Boundary improved model fit, both when it came to predicting Log RT [$X^2(1) = 56.85, p < .001$] and Correct Response [$X^2(1) = 83.29$, $p < .001$]. In other words, the distinction between Same Sense and Different Sense uses was predictive of both measures of processing difficulty, above and beyond three distinct measures of the continuous distance between prime and target uses.

Discussion

Together, these analyses suggest several conclusions.

First, it appears that Cosine Distance from each NLM is indeed predictive of RT. This means that regardless of the differences in model architecture, each NLM is capturing regularities in distributional statistics that allow them to produce separable contextualized embeddings, whose distance in state-space reflect degrees of processing difficulty on the task. Interestingly, BERT-base was the most predictive of RT, despite having fewer layers and fewer parameters than BERT-large. On the other hand, BERT-large was more predictive of Correct Response. It is unclear why this difference arises: one possibility is that BERT-large actually produces more separation between distinct senses of a wordform, thus leading to a sharper, categorical differences in Cosine Distance—which may be more amenable to predicting a binary outcome like Correct Response than a continuous one like RT. However, this conclusion is speculative, and merits future exploration.

Second, Sense Boundary remained predictive of human behavior, even when considering Cosine Distance from all three NLMs combined. This represents a particularly stringent test of Sense Boundary—i.e., whether it explains variance above and beyond the three NLMs together––and so the fact that it remains predictive suggests that the result is quite robust. Theoretically, this result is important because it echoes the earlier disconfirmation of the Pure Exemplar Theory, which predicts no effect of Sense Boundary. Even when considering ELMo, BERT-base, and BERT-large together, Sense Boundary explains independent variance in human behavior. This reinforces confidence in the conclusion that Hybrid Meaning Theory is the best explanation of the experimental data.

Supplementary Analysis 2: Sense Dominance

Past work has found that the difficulty transitioning between meanings of an ambiguous word depends on some extent to the relative *dominance* of those meanings (Duffy et al, 1988; Klepousniotou et al, 2008; Foraker & Murphy, 2012; Blott et al, 2020). Specifically, evidence suggests that it is easier to transition from a *subordinate* (i.e., less frequent) meaning to a *dominant* (more frequent) meaning than the reverse. Our original, pre-registered analyses did not incorporate a role for sense dominance. Thus, we conducted a post-hoc, exploratory analysis to ask: 1) whether accounting for sense dominance changed any of the substantive results; and 2) whether we could replicate the effect of dominance previously observed.

Obtaining dominance judgments

We first conducted a norming study with 79 participants; one was removed for failing a bot check, resulting in a total of 78 (53 female, 23 male, 1 non-binary, and 1 who preferred not to answer). The average age was 20.73 (SD = 2.24), and ranged from 18 to 35.

As in the relatedness norming study, participants were presented with a series of sentence pairs, with the target word **bolded** in each sentence (e.g., "They liked the marinated **lamb**"). The sentences were presented next to each other, i.e., on the right and left side of the screen. Only Different Sense pairs were presented. Participants were asked to rate the dominance of the target word's meaning in the sentence on the right side of the screen, on a 4-point scale ranging from -2 (left side is much more dominant) to 2 (right side is much more dominant).

For each sentence pair in a particular order, we averaged across different participants' ratings to create a Dominance Score. In the primary experiment, this Dominance Score corresponded to the relative dominance of the meaning in the target sentence compared to the meaning in the prime sentence: a more positive score indicated that the target sentence's meaning was more dominant, while a more negative score indicated that the prime sentence's meaning was more dominant.

Analysis and Results

We pooled the data from both Experiment 1 and Experiment 2, resulting in a total of 19,828 observations. As Dominance Scores were only available for Different Sense items, we further restricted these observations to be Different Sense items (13,227 observations), and merged them with the dominance data.

We then carried out a similar set of analyses as described in the primary manuscript, i.e., considering both Log RT and Correct Response as dependent variables. The primary difference was that Sense Boundary was no longer included in the models, given that observations included only Different Sense Items. Thus, the full model for each dependent variable included fixed effects of Dominance, Cosine Distance, and Ambiguity Type, as well as by-subject random slopes for each effect, and random intercepts for subjects, items, and experiment. The analysis of Log RT considered only trials on which the response was correct.

First, the full model explained significantly more variance than a model omitting only an effect of Dominance, for both Log RT [$X^2(1) = 9.29, p = .002$] and Correct Response [$X^2(1) = 185.39, p < .001$]. Trials on which the target item was more dominant were more likely to be responded to correctly [$\beta = 0.52, SE = 0.04$], and also had faster response times [$\beta = -0.04, SE = 0.004$].

Importantly, as in the primary analyses, Cosine Distance also significantly improved the fit of a model predicting Log RT [$X^2(1) = 23.22$, $p < .001$]. Items with a larger Cosine Distance between the prime and target contextualized embedding had slower response times [$\beta = 0.18$, SE

= 0.04]. This indicates that the effect of continuity holds even when considering Different Sense items only, and when controlling for Dominance.

Finally, we detected a main effect of Ambiguity Type for both Correct Response [$X^2(1) = 7.25, p = .007$] and Log RT [$X^2(1) = 6.29, p = .01$]. Polysemous target items were more likely to be answered correctly [β = 0.36, SE = 0.13] and had faster response times on average [β = -0.04, SE = 0.02]. Note that this last finding is consistent with the maximal hybrid account, i.e., the Hybrid+ Theory. However, it is also impossible to rule out the possibility that this main effect was driven by uncontrolled differences between the stimuli (i.e., between polysemous and homonymous items). Critically, this possibility could be disconfirmed by finding an *interaction* between Sense Boundary and Ambiguity Type, which we did not detect in either Experiment 1 or Experiment 2.

Overall, this exploratory analysis is reassuring for two reasons. First, it confirms that the main effect of Cosine Distance is detected even when considering only Different Sense items, and after controlling for Dominance. This strengthens the conclusion that the Mental Dictionary Framework should be rejected. And second, it replicates effect of sense dominance observed in past work: participants were faster, and more accurate, when the target meaning was more dominant (as rated by a separate pool of human participants) than the prime meaning.

Supplementary Analysis 3: Surprisal

In our pre-registered analyses, we found that including a fixed effect of Sense Boundary improved the explanatory of a model, above and beyond Cosine Distance (a measure of the continuous distance between the prime and target contexts of use). We also found that Cosine Distance explained additional variance beyond Sense Boundary. Together, we interpreted these results as disconfirming Pure Exemplar Theory (as well as both theories in the Mental Dictionary Framework) and providing support for Hybrid Meaning Theory.

One concern with this conclusion is whether Cosine Distance is the appropriate measure to extract from BERT. One popular alternative measure is surprisal: the negative log probability that a model assigns to a target word (or series of words). Surprisal has been used to predict a number of psycholinguistic measures, including reading time (Monsalve et al, 2012; Goodkind & Bicknell, 2018), the size of the N400 effect (Frank et al, 2015; Michaelov et al, 2020), and eye movements during reading (Kuribayashi et al, 2021). This raises the question of whether Surprisal would explain substantive variance on either of the dependent variables measured in our task (Accuracy and RT)—and critically, whether categorical factors like Sense Boundary would continue to explain additional variance in those measures after accounting for Surprisal.

Calculating Surprisal

We used BERT (large) to calculate Surprisal. Specifically, we presented BERT with each unique combination of sentences (accounting for order), to mimic the structure of the experiment (e.g., “They liked the marinated lamb. They liked the friendly lamb.”). We “masked” the critical word in the target sentence (e.g., “They liked the friendly [MASK]”), and measured the probability BERT assigned to that target word. In other words, we measured the likelihood of the target word appearing in that context, after the previous sentential context. We then calculated Surprisal by taking the negative log of this probability.

Analysis and Results

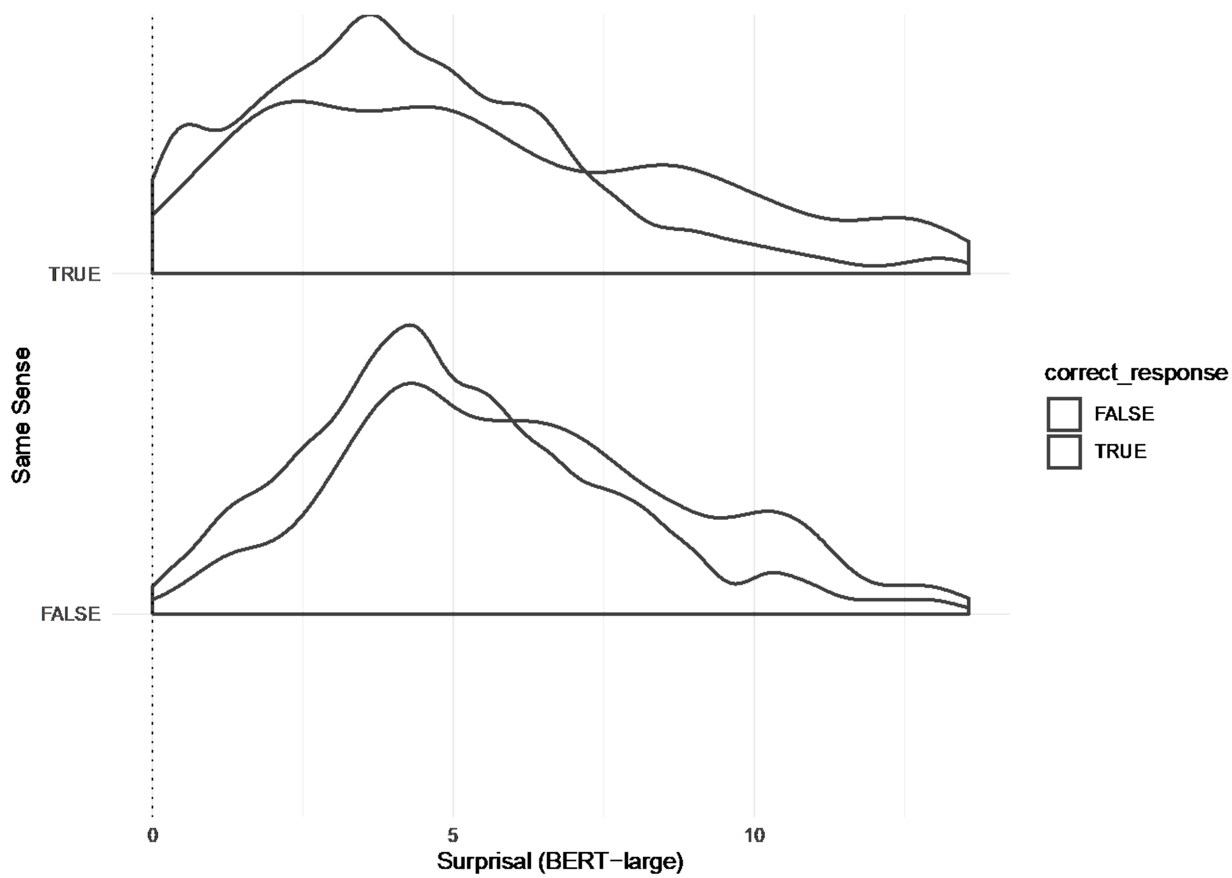
Our primary question was whether a fixed effect Sense Boundary continued to explain additional variance beyond a model omitting only that effect. Thus, for both Accuracy and RT, we constructed a full model including fixed effects of Surprisal, Cosine Distance, Sense Boundary, and Prior Accuracy (or Prior RT), and random intercepts for subjects and words. We compared this model to a reduced model omitting only Sense Boundary. Critically, the inclusion of Sense Boundary improved model fit for both Accuracy [$X^2(1) = 188.66, p < .001$] and RT [$X^2(1) = 71.58, p < .001$]. This indicates that even when Surprisal is accounted for (in addition to Cosine Distance), Sense Boundary continues to explain additional variance—providing support for Hybrid Meaning Theory.

We also asked whether Surprisal explained independent variance from Cosine Distance and Sense Boundary. We found that the full model explained significantly more variance than a reduced model omitting only Surprisal, for both Accuracy [$X^2(1) = 13.57, p = .001$] and RT [$X^2(1) = 27.36, p < .001$]. In both cases, higher Surprisal was associated with increased processing difficulty, i.e., lower Accuracy [$\beta = -0.05$, SE = 0.01] and longer RT [$\beta = 0.19$, SE = 0.02].

Discussion

The primary motivation for this supplementary analysis was to ask whether Hybrid Meaning Theory continued to be the best explanation of the data, even when using a different measure from BERT. The answer appears to be yes: for both RT and Accuracy, the existence of a sense boundary explains variance beyond both continuous measures tested (Cosine Distance and Surprisal). In other words, Pure Exemplar Theory is insufficient to explain the behavioral data.

An unexpected finding from the supplementary analysis was that Surprisal explained variance in *both* Accuracy and RT (while Cosine Distance explained variance only in RT). This is illustrated in *Supplementary Figure 3* below.



Supplementary Figure 3: For both Same Sense and Different Sense prime/target pairs, incorrect responses were associated with higher Surprisal values from BERT-large than correct responses. In other words, higher Surprisal on the critical word was associated with an incorrect response.

Supplementary Analysis 4: Contextualized Sensorimotor Norms

In the primary manuscript, we found that variation in RT and Accuracy was systematically predicted by the existence of a Sense Boundary—above and beyond Cosine Distance. However, as noted in the General Discussion, one concern with this analysis is that the measure of contextual distance was derived from BERT, which is exposed to linguistic input alone. It is possible that the variation currently attributed to Sense Boundary could actually be due to variation in some unmeasured variable, such as distance in a continuous *sensorimotor* space—which would not be accounted for by BERT.

In this analysis, we attempted to address this limitation using the Contextualized Sensorimotor Norms, or CS Norms (Trott & Bergen, 2022). The CS Norms include judgments about the strength or salience of various sensorimotor features (e.g., visual strength, auditory strength) of a word in different linguistic contexts. They are modeled on the Lancaster Sensorimotor Norms (Lynott et al., 2019), which includes judgments about the strength of these sensorimotor dimensions for approximately 40K English words, but which, crucially, does not account for potential variation in sensorimotor strength across contexts; for example, "marinated lamb" might emphasize the olfactory properties of lamb, while "furry lamb" might emphasize the tactile properties. The CS Norms were created to address that gap. Importantly, they allow us to calculate the *sensorimotor distance* between two contexts of use: each word/context pairing is represented by an 11-dimensional vector, and so the sensorimotor overlap between these contexts can be calculated by taking the cosine distance between these vectors (as with BERT embeddings).

Our approach was as follows. In addition to including the measure of Cosine Distance derived from BERT, we included another measure derived from the CS Norms. We then asked whether Sense Boundary continued to explain variance in the dependent measures used (RT and Correct Response), above and beyond *both* continuous measures. If Sense Boundary no longer explains additional variance, it would suggest that the variance previously attributed to a categorical variable (Sense Boundary) may actually have been attributable to another unmeasured continuous variable. If, however, Sense Boundary does continue to explain variance, it suggests that word meaning is indeed still at least partially categorical, even accounting for two distinct operationalizations of continuous distance.

Crucially, Sense Boundary continued to explain additional variance in both dependent variables, above and beyond fixed effects of Sensorimotor Distance (i.e., from the CS Norms) and Distributional Distance (i.e., from BERT): Log RT [$X^2(1) = 70.56$, $p < .001$] and Correct Response [$X^2(1) = 193.71$, $p < .001$]. (Note that Sensorimotor Distance also explained distinct variance in Log RT above and beyond the covariates tested, [$X^2(1) = 6.42$, $p = .01$], suggesting that performance on the task was also influenced by sensorimotor overlap, beyond distributional similarity or the existence of a sense boundary.)

Thus, it is unlikely that the results obtained in the pre-registered analyses are due to unmeasured variation in which sensorimotor dimensions are most present. Of course, a key limitation of this analyses (and this dataset) is that it does not account for differences in the *content* of sensorimotor experience, but rather differences in which modalities are more or less active in a given context. For example, the concepts of jasmine and ammonia might both be very correlated with olfactory sensation, but the specific olfactory experiences they involve are very different.

Of course, it is possible that alternative representations of semantic distance could explain away the effect of Sense Boundary. However, these results indicate that the effect is robust to the inclusion of at least two conceptually distinct covariates.